Часть 4

8.4.12. $\int \frac{dx}{x + \sqrt[3]{x^2}} = \int \frac{dx}{(\sqrt{x^3})^2 + \sqrt[3]{x^2}} = \Big[$ 4 случай ① $[x = t^k,\ k = НОК(2;3) = 6 \Rightarrow x = t^6$

$\Rightarrow dx = d(t^6) = (t^6)'_t\, dt = 6t^5 dt\Big] =$

$= \int \frac{6t^5 dt}{t^6 + \sqrt[3]{(t^6)^2}} = \int \frac{6t^5 dt}{t^6 + t^4} = \Big[\sqrt[3]{(t^6)^2} = t^{\frac{6\cdot 2}{3}} = t^{\frac{12}{3}} = t^4\Big] =$

$= 6\int \frac{t^5 dt}{t^6 + t^4} = 6\int \frac{t^5 dt}{t^4(t^2+1)} = 6\int \frac{t\, dt}{t^2+1} =$

$= \Big[y = t^2 + 1 \Rightarrow dy = d(t^2+1) = (t^2+1)'_t\, dt = 2t\, dt\Big] =$

$= 6\int \frac{t}{t^2+1} \cdot \frac{1}{2t} dy = 6\int \frac{1}{2y} dy = 3\int \frac{1}{y} dy =$

$= 3 \ln|y| + C = 3\ln|t^2+1| + C = \Big[x = t^6 \Rightarrow t = \sqrt[6]{x} \Rightarrow$

$\Rightarrow t^2 = (\sqrt[6]{x})^2 = \sqrt[3]{x}\Big] = 3\ln|\sqrt[3]{x} + 1| + C$

8.4.13 $\int \frac{\sqrt{x}}{1 + \sqrt[4]{x^3}} dx = \Big[$ 4 случай ① $k = НОК(2;4) = 4 \Rightarrow x = t^4 \Rightarrow$

$\Rightarrow dx = d(t^4) = (t^4)'_t\, dt = 4t^3 dt\Big] =$

$= \int \frac{4t^3 \cdot \sqrt{t^4}\, dt}{1 + \sqrt[4]{(t^4)^3}} =$

$= \int \frac{4t^3 \cdot t^2}{1 + t^3} dt = 4\int \frac{t^5}{t^3+1} dt = \Big[y = t^3 + 1 \Rightarrow$

$\Rightarrow dy = d(t^3+1) = (t^3+1)'_t\, dt = 3t^2 dt \Rightarrow dt = \frac{1}{3t^2} dy\Big] =$

$= 4\int \frac{t^5}{t^3+1} \cdot \frac{1}{3t^2} dy = 4\int \frac{t^3 dy}{3(t^3+1)} = \frac{4}{3}\int \frac{t^3+1-1}{y} dy =$

$= \frac{4}{3}\int \frac{y-1}{y} dy = \frac{4}{3}\int \frac{y}{y} dy - \frac{4}{3}\int \frac{1}{y} dy =$

$= \frac{4}{3}y - \frac{4}{3}\ln|y| + C = \frac{4}{3}(t^3+1) - \frac{4}{3}\ln|t^3+1| + C =$

$= [x = t^4 \Rightarrow t = \sqrt[4]{x} \Rightarrow t^3 = \sqrt[4]{x^3}] = \frac{4}{3}(\sqrt[4]{x^3}+1) -$

$- \frac{4}{3}\ln|\sqrt[4]{x^3}+1| + = \frac{4}{3}\sqrt[4]{x^3} - \frac{4}{3}\ln|\sqrt[4]{x^3}+1| + C$

8.4.14

8 случай ①

$\int \frac{x + \sqrt[3]{x^2} + \sqrt[6]{x}}{x(1-\sqrt[3]{x})} dx = [k = НОК(3,6) = 6 \Rightarrow x = t^6 \Rightarrow dx = d(t^6) =$

$= 6t^5 dt] = \int \frac{t^6+t^4+t}{t^6(1-t^2)} 6t^5 dt =$

$= 6\int \frac{t(t^5+t^3+1)}{t(1-t^2)} dt = 6\int \frac{t^5+t^3+1}{1-t^2} dt =$

$= -6\int \frac{t^5+t^3+1}{t^2-1} dt = \left[\begin{array}{l|l} t^5+0\cdot t^4+t^3+0\cdot t^2+0\cdot t+1 & t^2-1 \\ \hline t^5+0+t^3 & t^3+2t \\ -2t^3 & \\ 2t+0-2t & \\ 2t+1 & \end{array} \right]$

$= -6\int \frac{2t+1}{t^2-1} dt - 6\int t^3 dt - 12\int t\, dt =$

$= [\ 1)\ \int \frac{2t+1}{t^2-1} dt = 2\int \frac{t}{t^2-1} dt + \int \frac{1}{t^2-1} dt] =$

$= -12\int \frac{t\,dt}{t^2-1} - 6\int \frac{dt}{t^2-1} - 6\int t^3 dt - 12\int t\, dt =$

$= [\ 1)\ y = t^2-1 \Rightarrow dy = d(t^2-1) = 2t\,dt \Rightarrow dt = \frac{1}{2t}dy] =$

$= -12\int \frac{1}{2y} dy - 6\int \frac{dt}{t^2-1} - 6\int t^3 dt - 12\int t\, dt =$

$= -6\ln|y| - 3\cdot\ln\left|\frac{t-1}{t+1}\right| - 6\frac{t^4}{4} - 12\frac{t^2}{2} + C =$

$= -6\ln|t^2-1| - 3\ln\left|\frac{t-1}{t+1}\right| - \frac{3}{2}t^4 - 6t^2 + C =$

$= -6\ln|\sqrt[3]{x}-1| - 3\ln\left|\frac{\sqrt[6]{x}-1}{\sqrt[6]{x}+1}\right| - \frac{3}{2}\sqrt[3]{x^2} - 6\sqrt[3]{x} + C$

8.4.15

x случай ①

$\int \frac{\sqrt{x}\,dx}{x-\sqrt[3]{x^2}} = [k = \text{НОК}(2;3) = 6 \Rightarrow x = t^6 \Rightarrow dx = d(t^6) = 6t^5dt] =$

$= \int \frac{\sqrt{t^6}}{t^6-\sqrt[3]{t^{12}}}\cdot 6t^5dt = 6\int\frac{t^8}{t^6-t^4} = 6\int\frac{t^8}{t^4(t^2-1)}dt =$

$= 6\int\frac{t^4}{t^2-1}dt = \left[\begin{array}{l|l} t^4+0\cdot t^3+0\cdot t^2+0t+0 & t^2-1 \\ \underline{t^4+0t^3-t^2} & t^2+1 \\ \quad -t^2+0+0 & \\ \quad \underline{t^2+0-1} & \\ \quad\quad 1 & \end{array}\right] =$

$= 6\int\frac{(t^2-1)(t^2+1)+1}{t^2-1}dt = 6\int\frac{dt}{t^2-1} + 6\int t^2dt + 6\int dt =$

$= 6\cdot\frac{1}{2}\cdot\ln\left|\frac{t-1}{t+1}\right| + 6\frac{t^3}{3} + 6t + C = 3\ln\left|\frac{\sqrt[6]{x}-1}{\sqrt[6]{x}+1}\right| + 2\sqrt{x} + 6\sqrt[6]{x} + C$

8.4.16

x случай ①

$\int\frac{\sqrt{x}}{1+\sqrt{x}}dx = [k=2 \Rightarrow x = t^2 \Rightarrow dx = d(t^2) = 2tdt] =$

$= \int\frac{\sqrt{t^2}}{1+\sqrt{t^2}}dt = \int\frac{t}{1+t}\cdot 2tdt = 2\int\frac{t^2}{t+1}dt =$

$= 2\int\frac{t^2-1+1}{t+1}dt = 2\int\frac{(t-1)(t+1)}{t+1}dt + 2\int\frac{dt}{t+1} =$

$= 2\int tdt - 2\int dt + 2\int\frac{dt}{t+1} = 2\frac{t^2}{2} - 2t + 2\ln|t+1| + C =$

$= x - 2\sqrt{x} + 2\ln|\sqrt{x}+1| + C$

8.4.17

I случай ①

$$\int \frac{\sqrt{x}}{1+\sqrt[4]{x^3}}\,dx = \left[\begin{matrix} k = НОК(2,4)=4 \Rightarrow x=t^4 \\ dx = d(t^4) = 4t^3\,dt \end{matrix}\right] =$$

$$= \int \frac{\sqrt{t^4}\cdot 4t^3}{1+\sqrt[4]{t^{12}}}\,dt = \int \frac{t^2\cdot 4t^3}{1+t^3}\,dt = 4\int \frac{t^5}{t^3+1}\,dt =$$

$$= \left[y=t^3+1 \Rightarrow dy = d(t^3+1) = 3t^2\,dt \quad dt = \frac{1}{3t^2}\,dy\right] =$$

$$= 4\int \frac{t^5}{t^3+1}\cdot\frac{1}{3t^2}\,dy = \frac{4}{3}\int\frac{t^3+1-1}{t^3+1}\,dy =$$

$$= \frac{4}{3}\int\frac{y-1}{y}\,dy = \frac{4}{3}\int\frac{y}{y}\,dy - \frac{4}{3}\int\frac{1}{y}\,dy = \frac{4}{3}y - \frac{4}{3}\ln|y|+C =$$

$$= \frac{4}{3}(t^3+1) - \frac{4}{3}\ln|t^3+1|+C = \frac{4}{3}t^3 + \frac{4}{3} - \frac{4}{3}\ln|t^3+1|+C =$$

$$= \left[x=t^4 \Rightarrow t=\sqrt[4]{x} \quad t^3=\sqrt[4]{x^3}\right] =$$

$$= \frac{4}{3}\sqrt[4]{x^3} - \frac{4}{3}\ln|\sqrt[4]{x^3}+1|+C = \frac{4}{3}\sqrt[4]{x^3} - \frac{4}{3}\ln|\sqrt[4]{x^3}+1|+C$$

8.4.18

I случай ②

$$\int\frac{\sqrt{x+2}}{x}\,dx = \left[\, ax+b = t^k,\ k = НОК(n,q,s) \Rightarrow k=2 \Rightarrow\right.$$

$$\left.\Rightarrow x+2=t^2 \Rightarrow x=t^2-2;\ dx = d(t^2-2) = 2t\,dt\right] =$$

$$= \int\frac{\sqrt{t^2-2+2}}{t^2-2}\cdot 2t\,dt = \int\frac{2t^2}{t^2-2}\,dt = 2\int\frac{t^2-2}{t^2-2}\,dt +$$

$$+ 2\int\frac{2}{t^2-2}\,dt = 2\int dt + 4\int\frac{dt}{t^2-2} = 2t + 4\,\frac{1}{2\sqrt{2}}\ln\left|\frac{t-\sqrt{2}}{t+\sqrt{2}}\right|+C =$$

$$= 2\sqrt{x+2} + \sqrt{2}\ln\left|\frac{\sqrt{x+2}-\sqrt{2}}{\sqrt{x+2}+\sqrt{2}}\right|+C$$

8.4.19

$$\int \frac{x\,dx}{\sqrt{x+1}+\sqrt[3]{x+1}} = \left[\begin{array}{l} x \text{ случ. } 2 \\ k = \text{НОК}(2;3) = 6 \Rightarrow x+1 = t^6,\ x = t^6-1,\ dx = d(t^6-1) \Rightarrow \\ \Rightarrow 6t^5\,dt \end{array}\right] = \int \frac{t^6-1}{\sqrt{t^6}+\sqrt[3]{t^6}}\cdot 6t^5\,dt = 6\int \frac{(t^6-1)t^5}{t^3+t^2}\,dt =$$

$$6\int \frac{t^9-t^3}{t+1}\,dt = \left[\begin{array}{l|l} 1t^9+0t^8+0t^7+0t^6+0t^5+0t^4-1t^3 & t+1 \\ \underline{t^9+t^8} & t^8-t^7+t^6-t^5+t^4-t^3 \\ \quad -t^8+0t^7 & \\ \quad \underline{-t^8-t^7} & \\ \qquad t^7+0t^6 & \\ \qquad \underline{t^7+t^6} & \\ \qquad\quad -t^6+0t^5 & \\ \qquad\quad \underline{-t^6-t^5} & \\ \qquad\qquad t^5+0t^4 & \\ \qquad\qquad \underline{t^5+t^4} & \\ \qquad\qquad\quad -t^4-1t^3 & \\ \qquad\qquad\quad \underline{-t^4-t^3} & \\ \qquad\qquad\qquad 0 & \end{array}\right] =$$

$$= 6\int \frac{(t+1)(t^8-t^7+t^6-t^5+t^4-t^3)}{t+1}\,dt = 6\left(\int t^8\,dt - \int t^7\,dt + \right.$$

$$\left. + \int t^6\,dt - \int t^5\,dt + \int t^4\,dt - \int t^3\,dt\right) =$$

$$= 6\left(\frac{t^9}{9} - \frac{t^8}{8} + \frac{t^7}{7} - \frac{t^6}{6} + \frac{t^5}{5} - \frac{t^4}{4}\right) + C =$$

$$= \frac{2}{3}\sqrt[2]{(x+1)^3} - \frac{3}{4}\sqrt[3]{(x+1)^4} + \frac{6}{7}\sqrt[6]{(x+1)^7} - (x+1) + \frac{6}{5}\sqrt[6]{(x+1)^5} -$$

$$- \frac{3}{2}\sqrt[3]{(x+1)^2} + C$$

8.4.20

$$\int \frac{dx}{(x+1)^{3/2}+(x+1)^{1/2}} = \int \frac{dx}{\sqrt{(x+1)^3}+\sqrt{x+1}} = \left[\begin{array}{l} x \text{ случ. } 2 \qquad k = 2,\ x+1 = t^2 \\ k = \text{НОК}(2;2) = 2 \qquad x = t^2-1 \\ dx = d(t^2-1) = 2t\,dt \end{array}\right] = \int \frac{2t\,dt}{t^3+t} = 2\int \frac{dt}{t^2+1} =$$

$$= 2\,\frac{1}{1}\,\text{arctg}\,\frac{t}{1} + C = 2\,\text{arctg}(\sqrt{x+1}) + C$$

8.4.21

$$\int \frac{\sqrt{1+x}+1}{\sqrt{1+x}-1}\,dx = \left[\begin{matrix} x \text{ сл. } ② & & dx = d(t^2-1) = 2t\,dt \\ K=2, & x+1=t^2, & t^2-1=x \end{matrix}\right] =$$

$$= \int \frac{t+1}{t-1}\,2t\,dt = 2\int \frac{t^2+t}{t-1}\,dt = \left[\begin{array}{r|l} t^2+t & t-1 \\ \underline{t^2-t} & t+2 \\ 2t+0 & \\ \underline{2t-2} & \\ 2 & \end{array}\right] =$$

$$= 2\int \frac{(t-1)(t+2)+2}{t-1}\,dt = 2\int t\,dt + 4\int dt + 4\int \frac{dt}{t-1} =$$

$$= t^2 + 4t + 4\ln|t-1| + C = x+1+4\sqrt{x+1} + 4\ln\left|\sqrt{x+1}-1\right| + C =$$

$$= x + 4\sqrt{x+1} + 4\ln\left|\sqrt{x+1}-1\right| + C$$

8.4.22

$$\int \frac{x-1}{\sqrt{2x-1}}\,dx = \left[\begin{matrix} x \text{ сл. } ② & & x = \frac{t^2+1}{2} \\ K=2, & 2x-1=t^2, & dx = d\left(\frac{1}{2}t^2+\frac{1}{2}\right) = t\,dt \end{matrix}\right] =$$

$$= \int \frac{\frac{t^2+1}{2}-1}{\sqrt{t^2}}\,t\,dt = \int \frac{\frac{1}{2}t^2+\frac{1}{2}-1}{t}\,t\,dt = \frac{1}{2}\int t^2-1\,dt =$$

$$= \frac{1}{2}\int t^2\,dt - \frac{1}{2}\int dt = \frac{1}{2}\cdot\frac{t^3}{3} - \frac{1}{2}t + C = \frac{\sqrt{(2x-1)^3}}{6} - \frac{1}{2}\sqrt{2x-1} + C =$$

$$= \frac{(2x-1)\sqrt{2x-1}}{6} - \frac{3\sqrt{2x-1}}{6} + C = \frac{1}{6}\sqrt{2x-1}\,(2x-4) + C =$$

$$= \frac{1}{3}\sqrt{2x-1}\,(x-2) + C$$

8.4.23

$$\int \frac{dx}{\sqrt{1-2x} - \sqrt[4]{1-2x}} dx = \left[\begin{matrix} x \text{ случ. } 2 \\ k = \text{НОК}(2;4) = 4 \end{matrix} \quad \begin{matrix} 1-2x = t^4 \\ x = \frac{1}{2} - \frac{t^4}{2} \end{matrix} \quad \begin{matrix} dx = d\left(-\frac{1}{2}t^4 + \frac{1}{2}\right) = \\ = -2t^3 dt \end{matrix} \right] =$$

$$= \int \frac{-2t^3 dt}{t^2 - t} = -2 \int \frac{t^3 dt}{t(t-1)} = -2 \int \frac{t^2 dt}{t-1} =$$

$$= \left[t^2 = t^2 - 1 + 1 = (t-1)(t+1) + 1 \right] = -2 \int t+1 \, dt - 2 \int \frac{dt}{t-1} =$$

$$= -2\int t - 2\int dt - 2\int \frac{dt}{t-1} = -t^2 - 2t - 2\ln|t-1| + C =$$

$$= -\sqrt{1-2x} - 2\sqrt[4]{1-2x} - 2\ln\left|\sqrt[4]{1-2x} - 1\right| + C$$

8.4.24

$$\int \frac{1}{(2-x)^2} \sqrt{\frac{2-x}{2+x}} dx = \left[\begin{matrix} x \text{ случ. } 3 \\ k = 2 \end{matrix} \quad \frac{2-x}{2+x} = t^2, \quad 2-x = t^2(2+x) = \right.$$

$$\left. ; 2-x = 2t^2 + xt^2; \; 2 - 2t^2 = t^2 x + x; \; x = \frac{2-2t^2}{t^2+1}, \; dx = d\left(\frac{2-t^2}{t^2+1}\right) = -\frac{8t}{(t^2+1)^2} dt \right]$$

$$= \int \frac{1}{\left(\frac{2-2t^2}{t^2+1}\right)^2} \sqrt{t^2} \cdot \frac{(-8t)}{(t^2+1)^2} dt = \int \frac{(t^2+1)^2}{16t^4} t \frac{(-8t)}{(t^2+1)^2} dt =$$

$$= \int \frac{-dt}{2t^2} = -\frac{1}{2} \int t^{-2} dt = -\frac{1}{2} \frac{t^{-1}}{-1} + C = \frac{1}{2t} + C = \frac{\sqrt{2+x}}{2\sqrt{2-x}} + C$$

8.4.25

$$\int \frac{dx}{\sqrt{(x-1)^3(x-2)}} = \int \frac{\sqrt{x-1}}{\sqrt{(x-1)^4}} \frac{dx}{\sqrt{x-2}} = \int \frac{dx}{(x-1)^2 \sqrt{\frac{x-2}{x-1}}} = \left[\begin{matrix} x \text{ случ. } 3 \\ k = 2 \end{matrix} \quad \frac{x-2}{x-1} = t^2, \right.$$

$$\left. x - 2 = t^2 x - t^2, \; t^2 - 2 = x(t^2 - 1), \; x = \frac{t^2-2}{t^2-1}, \; dx = d\left(\frac{t^2-2}{t^2-1}\right) = \frac{2t}{t(t^2-1)^2} \right] =$$

$$= \int \frac{1}{\left(\frac{t^2-2}{t^2-1}-1\right)^2 t} \frac{2t}{(t^2-1)^2} dt = \left[\frac{t^2-2}{t^2-1} - 1 = \frac{t^2-2-t^2+1}{t^2-1} = \frac{-1}{t^2-1} \right] =$$

$$= \int \frac{(t^2-1)^2}{t} \frac{2t}{(t^2-1)^2} dt = \int 2 dt = 2t + C = 2\sqrt{\frac{x-2}{x-1}} + C$$

8.4.26

$$\int \frac{dx}{\sqrt[3]{(x-1)^2 (x+1)}} = \frac{dx}{(x-1)\sqrt[3]{\frac{x+1}{x-1}}} = \left[\begin{array}{l} A \text{ случ } 3 \quad x+1 = t^3 x - t^3 \\ k=3, \ \frac{x+1}{x-1} = t^3, \quad t^3+1 = x(t^3-1) \end{array} \right.$$

$$\left. x = \frac{t^3+1}{t^3-1}, \ dx = d\left(\frac{t^3+1}{t^3-1}\right) = -\frac{6t^2}{(t^3-1)^2} dt \right] =$$

$$= \int \frac{1}{\left(\frac{t^3+1}{t^3-1} - 1\right) t} \cdot \frac{-6t^2}{(t^3-1)^2} dt = -3 \int \frac{t}{t^3-1} dt =$$

$$= -3 \int \frac{t}{(t-1)(t^2+t+1)} dt = \left[\frac{t}{(t-1)(t^2+t+1)} = \frac{A}{t-1} + \frac{Bt+C}{t^2+t+1} \right.$$

$$t = At^2 + At + A + Bt^2 + Ct - Bt - C$$

$$t = (A+B)t^2 + (A+C-B)t + (A-C)$$

$$\begin{cases} A - C = 0 \\ A + C - B = 1 \\ A + B = 0 \end{cases} \quad \begin{cases} C = A - 0 \\ A + A - 0 - 0 + A = 1 \\ B = 0 - A \end{cases} \quad \left.\begin{cases} C = 1/3 \\ A = 1/3 \\ B = -1/3 \end{cases} \right] = -3 \int \frac{1}{3(t-1)} dt -$$

$$3 \int \frac{-t+1}{3(t^2+t+1)} dt = -\int \frac{dt}{t-1} + \int \frac{t-1}{t^2+t+1} dt =$$

$$= [2) \ A = 1, \ B = -1, \ q = 1, \ p = 1] = -\int \frac{dt}{t-1} + \frac{1}{2} \int \frac{(2t+1)dt}{t^2+t+1} + \left(-1 - \frac{1}{2}\right)$$

$$\int \frac{dt}{t^2+t+1} = \left[\begin{array}{l} 2) \ u = t^2 + t + 1, \ du = (2t+1) dt \\ 3) \ y = t + \frac{1}{2} \Rightarrow dy = dt \end{array} \right] =$$

$$= -\int \frac{dt}{t-1} + \frac{1}{2} \int \frac{du}{u} - \frac{3}{2} \int \frac{dy}{y^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = -\ln|t-1| + \frac{1}{2} \ln|u| -$$

$-\frac{3}{2}\cdot\frac{1}{\sqrt{3}/2}\cdot \operatorname{arctg}\frac{2y+1}{\sqrt{3}}+C=-\ln\left|\sqrt[3]{\frac{x+1}{x-1}}-1\right|+\frac{1}{2}\ln\left|\sqrt[3]{\left(\frac{x+1}{x-1}\right)^2}+\sqrt[3]{\frac{x+1}{x-1}}+1\right|-$

$-\sqrt{3}\operatorname{arctg}\left(\frac{2\sqrt[3]{\frac{x+1}{x-1}}+1}{\sqrt{3}}\right)+C$

8.4.27

$\int\frac{dx}{(1-x)\sqrt{1-x^2}}=\int\frac{dx}{(1-x)\sqrt{(1-x)(1+x)}}=\int\frac{dx}{(1-x^2)\sqrt{\frac{1+x}{1-x}}}=$

$=\left[k=2\Rightarrow \frac{1+x}{1-x}=t^2,\ 1+x=t^2-t^2x,\ 1-t^2=x(-t^2-1)\right.$

$\left.x=\frac{t^2-1}{t^2+1},\ dx=d\left(\frac{t^2-1}{t^2+1}\right)=\frac{4t}{(t^2+1)^2}dt\right]=$

$=\frac{1}{\left(1-\frac{t^2-1}{t^2+1}\right)t}\cdot\frac{4t\,dt}{(t^2+1)^2}=\int\frac{(t^2+1)^2}{4t}\,\frac{4t}{(t^2+1)^2}\,dt=$

$=\int dt=t+C=\sqrt{\frac{1+x}{1-x}}+C$

8.4.28

$\int\frac{dx}{x(1+\sqrt[3]{x})^3}=\int x^{-1}(1+x^{1/3})^{-3}dx=\left[m=-1,\ a=1,\ b=1,\ n=\frac{1}{3};\right.$

$\left.p=-3;\ 1)\ p\in\mathbb{Z}\Rightarrow k=3\Rightarrow x=t^3;\ dx=(t^3)'_t\,dt=3t^2dt\right]=$

$=\int t^{-3}(1+t)^{-3}\cdot 3t^3dt=3\int t^{-1}(t^2\cdot 3t^2+3t+1)^{-3}dt=$

$=3\int(t^4+3t^3+3t^2+t)^{-1}dt=-3\int\frac{dt}{t(t+1)^3}=\left[t(t+1)^3=0;\ t=0,\ t=-1\ \frac{t}{t(t+1)^3}\right.$

$=\frac{A}{t}+\frac{B}{t+1}+\frac{C}{(t+1)^2}+\frac{D}{(t+1)^3};\ 1=A(t+1)^3+Bt(t+2)^2+C(t+1)t+Dt$

$1=t^3(A+B)+t^2(3A+2B+C)+t(3A+B+C+D)+A$

$\begin{cases}A+B=0\\3A+2B+C=0\\3A+B+C+D=0\\A=1\end{cases}\quad\begin{cases}B=-1\\3-2+C=0\\3-1+C+D=0\\A=1\end{cases}\quad\begin{cases}B=-1\\C=-1\\2-1+D=0\\A=1\end{cases}\quad\left.\begin{cases}A=1\\B=-1\\C=-1\\D=-1\end{cases}\right]=$

$=3\int\frac{dt}{t}-3\int\frac{dt}{t+1}-3\int\frac{dt}{(t+1)^2}-3\int\frac{dt}{(t+1)^2}=\left[1)\ dt=d(t+1)\right]=$

$$= 3\ln|t| - 3\ln|t+1| - 3\frac{(t+1)^{-1}}{-1} - 3\frac{(t+1)^{-2}}{-2} + C =$$

$$= 3\ln|x^{1/3}| - 3\ln|x^{1/3}+1| + 3(x^{1/3}+1)^{-1} + \frac{3}{2}(x^{1/3}+1)^{-2} + C =$$

$$= \ln|x| - 3\ln|\sqrt[3]{x}+1| + \frac{3}{\sqrt[3]{x}+1} + \frac{3}{2(\sqrt[3]{x}+1)^2} + C$$

8.4.29

$$\int x^3\cdot\sqrt{1+x^2}\,dx = \int x^3\cdot(1+x^2)^{1/2}dx = [m=3,\ n=2,\ p=1/2,$$

$$1)\ p=1/2 \in Z;\ 2)\ \frac{m+1}{n} = \frac{3+1}{2} = 2 \in Z \Rightarrow 1+x^2=t^2;$$

$$x=\sqrt{t^2-1},\ dx = \frac{t}{\sqrt{t^2-1}}dt] = \int \sqrt{(t^2-1)^3}\cdot t\cdot\frac{t}{\sqrt{t^2-1}}dt =$$

$$= \int t^2(t^2-1)\,dt = \int t^4 - t^2\,dt = \int t^4\,dt - \int t^2\,dt =$$

$$= \frac{1}{5}t^5 - \frac{1}{3}t^3 + C = \frac{1}{5}(\sqrt{1+x^2})^5 - \frac{1}{3}(\sqrt{1+x^2})^3 + C$$

8.4.30

$$\int\frac{dx}{x^{11}\cdot\sqrt{x^4+1}} = \int x^{-11}\cdot(x^4+1)^{-1/2}dx = [m=-11,\ n=4,\ p=-1/2,$$

$$1)\ p=-\frac{1}{2} \in Z;\ 2)\ \frac{m+1}{n} = \frac{-11+1}{4} = -\frac{10}{4} \in Z\ \ 3)\ \frac{m+1}{n}+p =$$

$$= \frac{-11+1}{4} - \frac{1}{2} = +2{,}5-0{,}5 = 2 \in Z \Rightarrow x^{-4}+1=t^2\quad x=\sqrt[4]{\frac{1}{t^2-1}},\ dx = -\frac{t\,dt}{2\sqrt[4]{t^2-1}(t^2-1)}] =$$

$$= \int\sqrt[4]{(t^2-1)^4}\cdot\sqrt{\frac{t^2-1}{11}}(-1)\frac{t\,dt}{2\sqrt[4]{t^2-1}(t^2-1)} = \frac{1}{2}\int(t^2-1)^2\sqrt[4]{(t^2-1)^3}\cdot$$

$$\cdot\frac{\sqrt[4]{(t^2-1)^2}}{t}\cdot\frac{t\,dt}{\sqrt[4]{(t^2-1)^3}(t^2-1)} = \frac{1}{2}\int(t^2-1)^2dt = -\frac{1}{2}\int(t^4-2t^2+1)\,dt$$

$$= -\frac{1}{2}\int t^4\,dt + \int t^2\,dt - \int\frac{1}{2}\,dt = -\frac{1}{2}\cdot\frac{1}{5}\cdot t^5 + \frac{1}{3}\cdot t^3 - \frac{1}{2}t + C =$$

$$= -\frac{(x^4+1)^2\sqrt{x^4+1}}{10x^{10}} + \frac{(x^4+1)\sqrt{x^4+1}}{3x^6} - \frac{\sqrt{x^4+1}}{2x^2} + C =$$

$$= -\frac{(8x^8-4x^4+3)\sqrt{x^4+1}}{30x^{10}} + C$$

8.4.31

$$\int \frac{dx}{\sqrt{x}(1-\sqrt{x})^2} = \int x^{-1/2}(1-x^{1/2})^{-2}dx = [m=-1/2,\ n=1/2,\ p=-2,$$

$$1)\ p=-2 \in Z \Rightarrow k = \text{НОК}(2;2)=2;\ x=t^2,\ x^{1/2}=t,\ x^{-1/2}=t^{-1}$$

$$dx = 2t\,dt] = \int \frac{2t\,dt}{t(1-t)^2} = 2\int \frac{dt}{(t-1)^2} = [dt = d(t-1)] = 2\int (t-1)^{-2}\,d(t-1)$$

$$= \frac{2(t-1)^{-1}}{-1} + C = -\frac{2}{t-1} + C = \frac{2}{1-\sqrt{t}} + C$$

8.4.32

$$\int x^5 \sqrt[3]{(1+x^3)^2}\,dx = \int x^5(1+x^3)^{2/3}dx = [m=5,\ n=3,\ p=2/3,$$

$$1)\ p=2/3 \in Z,\ 2)\ \frac{m+1}{n} = \frac{5+1}{3} = 2 \in Z,\ k=3 \Rightarrow 1+x^3=t^3,\ x=\sqrt[3]{t^3-1}$$

$$dx = \frac{t^2}{\sqrt[3]{(t^3-1)^2}}dt] = \int \sqrt[3]{(t^3-1)^5}\cdot t^2 \cdot \frac{t^2}{\sqrt[3]{(t^3-1)^2}}dt = \int (t^3-1)t^4\,dt =$$

$$= \int t^7 dt - \int t^4 dt = \frac{1}{8}t^8 - \frac{1}{5}t^5 + C = \frac{1}{8}\sqrt[3]{(1+x^3)^8} - \frac{1}{5}\sqrt[3]{(1+x^3)^5} + C$$

8.4.33

$$\int \frac{dx}{x^3\cdot\sqrt[3]{2-x^3}} = \int x^{-3}(2-x^3)^{-1/3}dx = [m=-3,\ n=3,\ p=-1/3,$$

$$1)\ p=-1/3 \in Z,\ 2)\ \frac{m+1}{n} = \frac{-3+1}{3} \in Z,\ 3)\ \frac{m+1}{n}+p = \frac{-3+1}{3} - \frac{1}{3} = -\frac{2}{3} - \frac{1}{3} = -1$$

$$\in Z \Rightarrow k=3,\ 2x^{-3}-1 = t^3,\ x^{-3} = \frac{t^3+1}{2};\ x^3 = \frac{2}{t^3+1};\ x = \sqrt[3]{\frac{2}{t^3+1}}$$

$$dx = \frac{\sqrt[3]{2}\,t^2\,dt}{(t^3+1)\sqrt[3]{t^3+1}} = -\frac{1}{2}\int t\,dt = -\frac{1}{2}\frac{t^2}{2} + C = -\frac{t^2}{4} + C = -\frac{\sqrt[3]{(2x^{-3}-1)^2}}{4} + C$$

$$= -\frac{(x^3-2)^{2/3}}{4x^2} + C$$

8.4.34

$\int \sqrt{x}\,(1+\sqrt{x})^3 dx =$

$= \int \sqrt{x}\,(1+3\sqrt{x}+3x+x\sqrt{x})\,dx = \int(\sqrt{x}+3x+3x\sqrt{x}+x^2)\,dx =$

$= \int x^{1/2} dx + 3\int x\, dx + 3\int x^{3/2} dx + \int x^2 dx =$

$= \frac{x^{3/2}}{3/2} + 3\frac{x^2}{2} + 3\frac{x^{5/2}}{5/2} + x^3/3 + C =$

$= \frac{2}{3}x\sqrt{x} + \frac{3}{2}x^2 + \frac{6}{5}x^2\sqrt{x} + \frac{1}{3}x^3 + C$

8.4.35

$\int \sqrt{x^3-4}\, x^2 dx = [y = x^3-4,\ dy = d(x^3-4) = 3x^2 dx$

$dx = \frac{1}{3x^2}dy] = \int y x^2 \cdot \frac{1}{3x^2}dy = \int \frac{1}{3}\sqrt{y}\,dy = \frac{1}{3}\frac{2}{3}y^{3/2} + C =$

$= \frac{2}{9}(x^3-4)^{3/2} + C$

8.4.36

$\int \frac{dx}{\sqrt{1-2x-x^2}} = \frac{dx}{\sqrt{2-(x+1)^2}} = [p = x+1;\ dp = dx] = \int \frac{dp}{\sqrt{2-p^2}} =$

$= \arcsin\frac{p}{\sqrt{2}} + C = \arcsin\left(\frac{x+1}{\sqrt{2}}\right) + C = \arcsin\left(\frac{\sqrt{2}(x+1)}{2}\right) + C$

8.4.37

$\int \frac{(x-2)dx}{\sqrt{x^2-10x+29}} = \int \frac{(x+5)+3\,dx}{\sqrt{x^2-10x+25+4}} = \int \frac{x+5}{\sqrt{x^2-10+29}}dx + 3\int\frac{dx}{\sqrt{(x+5)^2+4}} =$

$= [1)\ t = x^2-10x+29 \Rightarrow dt = (2x-10)dx,\ \frac{1}{2}dt = (x-5)dx,$

$2)\ (x-5) = y,\ dx = dy] = \frac{1}{2}\int\frac{dt}{\sqrt{t}} + 3\int\frac{dy}{\sqrt{y^2+4}} = \frac{1}{2}\frac{t^{1/2}}{1/2} +$

$+ 3\ln|y+\sqrt{y^2+4}| + C = \sqrt{x^2-10x+29} + 3\ln|x-5+\sqrt{x^2-10x+29}| + C$

8.4.38

$$\int \frac{3x-5}{\sqrt{x^2-4x+5}}dx = \int \frac{3(x-2)+1}{\sqrt{x^2-4x+5}}dx = 3\int \frac{(x-2)\,dx}{\sqrt{x^2-4x+5}} +$$

$$+\int \frac{dx}{\sqrt{x^2-4x+5}} = [1)\ x^2-4x+5=t,\ dt=(2x-4)\,dx \Rightarrow$$

$$\Rightarrow \frac{1}{2}dt=(x-2)\,dx;\ 2)\ y=x-2,\ dx=dy] = 3\cdot\frac{1}{2}\int\frac{dt}{\sqrt{t}} +$$

$$+\int\frac{dy}{\sqrt{y^2+1}} = \frac{3}{2}\cdot\frac{t^{1/2}}{1/2} + \ln\left|y+\sqrt{y^2+1}\right| + C =$$

$$= 3\sqrt{x^2-4x+5} + \ln\left|x-2+\sqrt{x^2-4x+5}\right| + C$$

8.4.39

$$\int\frac{x+1}{\sqrt{2x-x^2}}dx = \int\frac{(x-1)+2}{\sqrt{2x-x^2}}dx = \int\frac{(x-1)\,dx}{\sqrt{2x-x^2}} +$$

$$+2\int\frac{dx}{\sqrt{2x-x^2}} = [1)\ 2x-x^2=t,\ dt=(2-2x)\,dx,\ \frac{1}{2}dt=(1-x)\,dx,$$

$$-\frac{1}{2}dt=(1-x)\,dx,\ -\frac{1}{2}dt=(x-1)\,dx;\ 2)\ y=x-1,\ 2x-x^2=1-1+2x-x^2=$$

$$=1-(x-1)^2=1-y^2,\ dx=dy] = -\frac{1}{2}\int\frac{dt}{\sqrt{t}} + 2\int\frac{dy}{\sqrt{1-y^2}} =$$

$$= -\frac{1}{2}\frac{t^{1/2}}{1/2} + 2\arcsin\frac{y}{1} + C = 2\arcsin(x-1) - \sqrt{2x-x^2} + C$$

8.4.40

$$\int\frac{\sqrt{1-x^2}}{x}dx = \int x^{-1}\sqrt{1-x^2}\,dx = [m=-1,\ n=2,\ p=1/2$$

$$1)\ p=1/2 \in Z \quad 2)\ \frac{m+1}{n} = \frac{-1+1}{2} = 0 \in Z \Rightarrow k=2,\ 1-x^2=t^2$$

$$x^2=1-t^2,\ x=\sqrt{1-t^2},\ dx=d\left(\sqrt{1-t^2}\right) = -\frac{t\,dt}{\sqrt{1-t^2}}] =$$

$$= \int\frac{t}{\sqrt{1-t^2}}(-1)\cdot\frac{t\,dt}{\sqrt{1-t^2}} = -\int\frac{t^2dt}{1-t^2} = \int\frac{t^2dt}{t^2-1} =$$

$$= [t^2=(t^2-1)+1] = \int\frac{(t^2-1)+1}{t^2-1}dt = \int dt + \int\frac{dt}{t^2-1} =$$

$$= t + \frac{1}{2\cdot 1}\ln\left|\frac{t-1}{t+1}\right| + C = \sqrt{1-x^2} + \frac{1}{2}\cdot\ln\left|\frac{\sqrt{1-x^2}-1}{\sqrt{1-x^2}+1}\right| + C$$

8.4.41

$\int \sqrt{4-x^2}\,dx = [x = 2\sin t,\ dx = 2\cos t\,dt] =$

$= \int \sqrt{4-(2\sin t)^2} \cdot 2\cos t\,dt = 2\int \sqrt{4-4\sin^2 t} \cdot \cos t\,dt =$

$= 4\int \sqrt{1-\sin^2 t}\,\cos t\,dt = 4\int \cos t \cdot \cos t\,dt =$

$= 4\int \cos^2 t\,dt = 4\int \frac{1-\cos 2t}{2}\,dt = 2\int (1+\cos 2t)\,dt =$

$= 2\int dt + 2\int \cos 2t\,dt = 2t + \sin 2t + C =$

$= 2\arcsin\frac{x}{2} + \sin\left(2\arcsin\frac{x}{2}\right) + C = 2\arcsin\frac{x}{2} + \frac{x\sqrt{4-x^2}}{2} + C$

8.4.42

$\int x \cdot \sqrt[5]{x-2}\,dx = \int x\,(x-2)^{1/5}\,dx = [m = 1,\ n = 1,\ p = 1/5,$

$1)\ p = 1/5 \in Z,\ 2)\ \frac{m+1}{n} = \frac{1+1}{1} = 2 \in Z \Rightarrow k = 5,\ x-2 = t^5,$

$x = t^5 + 2,\ dx = 5t^4\,dt] = \int (t^5+2)\,t \cdot 5t^4\,dt =$

$= 5\int t^{10}\,dt + 10\int t^5\,dt = \frac{5}{11}t^{11} + \frac{10}{6} \cdot t^6 + C =$

$= \frac{5}{11}\sqrt[5]{(x-2)^{11}} + \frac{5}{3}\sqrt[5]{(x-2)^6} + C$

8.4.43

$\int \frac{dx}{x^2\sqrt{x^2+1}} = \int x^{-2}(x^2+1)^{-1/2}\,dx = [m = -2,\ n = 2,\ p = -1/2,$

$1)\ p = -1/2 \in Z,\ 2)\ \frac{m+1}{n} = \frac{-2+1}{2} = -\frac{1}{2} \in Z$

$3)\ \frac{m+1}{n} + p = \frac{-2+1}{2} - \frac{1}{2} = -1 \in Z \Rightarrow k = 2 \Rightarrow x^{-2} + 1 = t^2$

$x^{-2} = t^2 - 1,\ x^2 = \frac{1}{t^2-1},\ x = \frac{1}{\sqrt{t^2-1}},\ dx = \frac{-t\,dt}{\sqrt{t^2-1}\,(t^2-1)}] =$

$= \int \frac{t^2-1}{\sqrt{\frac{1}{t^2-1}+1}}\,(-1)\frac{t\,dt}{\sqrt{t^2-1}\,(t^2-1)} = -\int dt = -t + C = -\sqrt{x^{-2}+1} + C$